MDS202 · MODULE 2 · ACTIVITIES 2 – 5

Data Wrangling with Pandas

Weeks 5 – 8 · Continue using the same approved Seattle dataset

WEEK	TOOLS	WEIGHT	FINAL OUTPUT
5 – 8	Pandas + NumPy	70% of Module 2 (15 / 15 / 20 / 20)	Notebook + cleaned CSV

Activity 2 Explore ▶ **Activity 3** Manipulate ▶ **Activity 4** Transform ▶ **Activity 5** Clean & Validate

Assessed in every activity

Besides each activity's own criterion, your work in every activity (Weeks 4–8) counts toward **Code Quality, Reproducibility & Documentation (5%)** and **Interpretation, Justification & Reflection (10%)**.

ACTIVITY 2 Data Exploration with Pandas

WEEK	WEIGHT	GOAL
5	15%	Understand the structure, content, and quality of your dataset before manipulating it

Load the dataset and produce a concise **data profile**:

- ☐ Display the first and last records
- ☐ Report the number of rows and columns
- ☐ List all columns and data types
- ☐ Separate numerical, categorical/text, and date/time variables
- ☐ Descriptive statistics for relevant numerical columns
- ☐ Identify columns with missing values
- ☐ Identify possible identifier columns
- ☐ Inspect unique values of key categorical columns

Data Dictionary

Document at least **8 important columns**.

Column	Data Type	Meaning	Potential Issue

Reflect

1. What are the most important variables in the dataset?
2. Which variables require additional investigation before analysis?
3. What data-quality issues did you identify during exploration?

ACTIVITY 3 Data Manipulation and Aggregation

WEEK	WEIGHT	GOAL
6	15%	Use Pandas to answer meaningful questions about the dataset

Develop **at least three analytical questions** that require data manipulation. At least **one** must use filtering and at least **two** must use `groupby()`.

YOUR WORK MUST DEMONSTRATE

- ☐ Sorting
- ☐ Boolean filtering
- ☐ At least one derived column
- ☐ `groupby()`
- ☐ At least three aggregation functions
- ☐ Appropriate handling of the index

ANSWER EVERY QUESTION USING THIS STRUCTURE

Question ▶ **Code** ▶ **Result** ▶ **Interpretation**

> **Do not simply report output**
> Explain what each result **means** for the dataset and its context.

Question Planner

#	Analytical question	Operations used	Key finding
Q1			
Q2			
Q3			

ACTIVITY 4 Data Transformation and Integration

WEEK	WEIGHT	GOAL
7	20%	Transform data into a form suitable for analysis and, where appropriate, integrate a second Seattle dataset

Part A — Transformation

Perform **at least three** appropriate transformations. **One of them must reshape the data**; choose the others from the list below.

> **Required — Reshaping**
> Reshape or summarize the data using an appropriate Pandas operation such as `pivot_table()`, `melt()`, or `stack()` / `unstack()`. Report `df.shape` **before and after**, clearly state what one row represents in the resulting table.

- ☐ Date/time extraction
- ☐ Unit conversion
- ☐ Renaming columns
- ☐ String standardization
- ☐ Binning / categorization
- ☐ Creating derived measures

For each transformation, explain its **analytical purpose**:

#	Transformation	Column(s) affected	Analytical purpose
T1	**Reshaping:**		
T2			
T3			

Part B — Integration

Select a second related Seattle Open Data dataset **only if a defensible relationship exists**. Identify the common key or relationship **before** merging. If no defensible dataset exists, explain in a Markdown cell which datasets you considered and why none were appropriate.

```
merged = pd.merge(df1, df2, on="COMMON_KEY", how="left")
```

Rows in df1	Rows in df2	Rows after merge	Duplicates introduced?	New missing values?

Reflect

1. Why is the second dataset relevant?
2. Why is the selected merge type appropriate?
3. What evidence shows that the merge worked correctly?

ACTIVITY 5 Data Cleaning and Validation

WEEK	WEIGHT	GOAL
8	20%	Produce a clean, validated, analysis-ready dataset while documenting every important decision

Decision Record

Investigate **each issue below** and complete one row per issue. Add rows for any other issues you find.

Issue	Evidence	Decision	Action (code)	Reason
Missing values				
Duplicate records				
Incorrect data types				
Inconsistent categorical values				
Invalid / impossible values				
Potential outliers				
Date/time inconsistencies				

Useful operations: `dropna()`, `fillna()`, `drop_duplicates()`, `astype()`, `pd.to_datetime()`, and the `.str` string-cleaning methods.

> **Important**
> Do not remove an observation merely because it is unusual. **Investigate and justify** the decision.

Validation Checklist

- ☐ Expected columns are present
- ☐ Important columns have appropriate data types
- ☐ Required identifiers are not unexpectedly duplicated
- ☐ Missing values have been investigated
- ☐ Categorical values are consistent
- ☐ Numerical values satisfy reasonable domain constraints
- ☐ Date/time values are valid
- ☐ The final row count is documented
- ☐ Cleaning decisions are reproducible from code
- ☐ The dataset can be used for a stated analysis purpose

Final Deliverable — Analysis-Ready Dataset

```
df.to_csv("seattle_cleaned_dataset.csv", index=False)
```

Submit

Your final **notebook** and the cleaned **CSV file**.

PORTFOLIO Final Reflection

1. Which data-wrangling step required the most reasoning rather than coding?
2. Which NumPy operation was most useful in your project?
3. Which Pandas operation was most useful?
4. What would be risky about analyzing the raw dataset without cleaning and validation?
5. What additional data would improve the analysis?
6. If this were a real organizational project, what would you document or automate next?

Submission Structure

Organize your final notebook in this order:

#	Notebook section	Source	#	Notebook section	Source
1	Dataset Selection	Selection Record	7	Integration (if applicable)	Activity 4B
2	Data Profile	Activity 2	8	Cleaning	Activity 5
3	NumPy Analysis	Activity 1	9	Validation	Activity 5
4	Pandas Exploration	Activity 2	10	Final Dataset	Activity 5
5	Manipulation & Aggregation	Activity 3	11	Findings	All
6	Transformation	Activity 4A	12	Reflection	Portfolio